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Performance analysis of vertical axis wind turbine using NACA0017 airfoil section

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ABSTRACT

The vertical axis wind turbine has many advantages as compared to horizontal axis wind turbine as they are much more convenient to use in urban areas, turbulent environment and low wind speed regions. However, its low efficiency and low self-start ability are the main concerns. For the performance enhancement of the turbine, selection of effective blade profile is an important criterion. In this paper two dimensional CFD simulation have been carried out on Darrieus lift type turbine having NACA0017 blade profile. The performance of this airfoil is compared with already established NACA0015 airfoil. The numerical results predicted that power coefficient of NACA0017 is relatively on higher side at higher TSR which also provides good structural strength due to its higher thickness. The peak value of power coefficient is found to be 0.4 and 0.38 at velocity 5m/s and 4m/s respectively.
Keywords: Vertical axis wind turbine; Airfoil shape; Performance enhancement; Power coefficient.

1. INTRODUCTION

In recent years, the demand of renewable energy has increased significantly due to the negative ill effects of depleted conventional fuels to the environment that led to air pollution, global warming, climate change etc. Among the alternative energy resources, wind energy has the potential to grow as an alternative source of energy which has very less adverse effect to the environment [1]. The wind energy conversion technology (WECT) is applied to extract maximum possible energy from the wind energy source. It has three stages: energy resource assessment, hardware design and installation. In the first stage, the wind power potential is evaluated for several parts of the world. For example, the wind power potential of East and Northeast region of India have been assessed by Rahman and Chattopadhyay by introducing a new technique to calculate the parameters involve in wind assessment [2]. The second stage is comprised of installation of an efficient wind turbine system. There are basically two types of wind turbine system. One is horizontal axis wind turbine (HAWT) and another is vertical axis wind turbine (VAWT). HAWT is widely used in the wind energy conversion technology as it has high power efficiency compared to VAWT despite it requires a complex yaw mechanism [3]. HAWTs also require high wind velocity and huge empty space to prevent aerodynamic interferences [4]. whereas, VAWTs can be installed in urban and semi urban areas due to its omni-directional characteristics, less noise, simple construction, low installation cost and maintenance [5]. Also, VAWTs can harness energy from highly turbulent wind flows which are very common in the urban areas [4]. VAWTs can also extract wind power from low wind speed regions as well as very high wind speed regions [6]. Recent research found that VAWT has better potentials for offshore platform compared to HAWTs in terms of scalability, simple design [7]. Again, VAWT can be classified into two groups according to the aerodynamics principles: lift type and drag type [8]. Darrieus VAWT is an example of lift type and Savonius VAWT is an example of drag type. H-type Darrieus wind turbine provides greater efficiency than drag type [9].

Three-dimensional solution of Navier-Stokes equation is used here to analyze VAWT as it provides results comparable to experimental results [10]. To cite some important work, Castelli et al. [11] assessed the aerodynamic performance of Darrieus wind turbine with three straight bladed having NACA0021 airfoils. The CFD analysis of a combined Darrieus and Savonius VAWT at various conditions is performed by Debnath et al.[12]. It showed that the CFD results were as good as experimental results. Nobile et al. [13] investigated a CFD simulation of augmented VAWT with NACA0018 blade profile. Combined experimental and CFD analysis of VAWT were presented by many authors [6, 14-17]. Similarly, several researchers have performed CFD simulations to study aerodynamic behavior of VAWT to enhance its performance [18-20].

The performance parameters such as number of blades, blade shape, pitch angle, wind speed, tip speed ratio (TSR), rotor solidity are need to be studied to evaluate the performance of VAWT. The performance of 3 bladed H-rotor Darrieus VAWT with 24 different shapes of airfoils have been analysed by Hashem and Mohamed [21]. The maximum power coefficient is provided by S1046 airfoil. Mohamed et al. [22] carried out a numerical analysis of H-type Darrieus VAWT using 25 airfoils shapes to study effects of airfoils shape and pitch angle. NACA0015 airfoil provided the best efficiency followed by NACA0018 out of NACA-4 digit series airfoils. 3D simulation has been performed on the performance analysis of VAWT using three different airfoils by Elkhoury et

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al. [23]. They used NACA0018, NACA0021 and NACA634-221 blade profiles. Both numerical and experimental results showed that NACA0021 has better power coefficient than the NACA0018, although NACA0018 is 3% thinner than the NACA0021.

The main objective of our present work is to assess the performance of H-type Darrieus VAWT with NACA0017 blade profile using numerical simulation. The effectiveness of this profile has not been studied till date [24]. The performance of this airfoil is compared with NACA0015 which has excellent performance in application of wind turbine.

2. COMPUTATIONAL MODELING

2.1 Flow domain and boundary conditions

The numerical analysis of the performance of an H-type Darrius VAWT with straight blade of NACA0017 air foil is conducted with the help of CFD software ANSYS-FLUENT 16.2 version. Three dimensional effects are neglected.

Earlier studies show that effect of domain size on the performance of the VAWT is significant [25]. The CFD analysis may give wrong results and there may be overestimation of the performance of VAWT if the computational domain is not selected properly. Rezaiha et al [25] showed guidelines for selecting minimum domain size and predicted an optimum domain size of 7.5D*22.5D where results are very less affected by uncertainty in the boundary conditions. The present computational domain has inlet and outlet distances from the centre of the rotor of 7.5D and 15D respectively where D is the rotor diameter. The domain size of 22.5D x 7.5D is shown in figure 1. The geometry has been divided in two regions: a rotating domain of circular interface and the rectangular stationary domain beyond the circular interface. The two regions are interconnected with each other by using sliding interface to maintain the continuity of the incoming velocity and to provide the correct values of velocities for the rotating domain.

The approaching velocity at the inlet boundary is taken as 5 m/s corresponding to a Reynolds number of 34,229. At the outlet, atmospheric pressure is used as the domain is connected with ambient and other surfaces are considered with vanishing derivatives of flow variables i.e., free-slip.

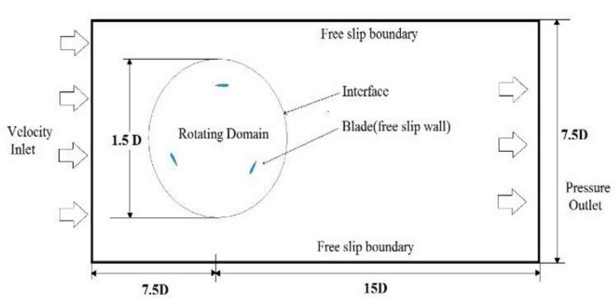

Figure 1: Computational model for present study

There are several parameters which affects the performance of VAWT such as number of blades, blade shape, pitch angle, wind speed, tip speed ratio (TSR), rotor solidity, torque coefficient (C_T), power coefficient (C_P) which are expressed in equation 1 to 5 [14]. To specify the rotor design, we have to consider an important parameter known as the rotor solidity (σ) of VAWT and it can be expressed as

$$\sigma = \frac{nc}{2R} \tag{1}$$

Where n, c and R are the number of blades, chord length of the blade and radius of the rotor respectively. The toque coefficient (C_T) and power coefficient (C_P) are used to evaluate the performance of wind turbine which are expressed as follows:

$$C_T = \frac{2T}{\rho A R v^2} \tag{2}$$

$$C_P = \frac{P}{\rho v^3 H R} \tag{3}$$

Where P is power and it is expressed as P=T.N/60, N is revolution per minute, T is torque in Nm, ρ is density of air, v is free stream velocity, A is cross sectional area of rotor.

C_P and C_T are related through the tip speed ratio (TSR) of airfoils as

$$C_P = C_T * TSR \tag{4}$$

Where $$TSR = \frac{\omega r}{v} \tag{5}$$

2.2 Meshing and model validation

Quad dominant mesh was selected for the whole computational domain. Mesh was refined near the airfoil walls with quadrilateral mesh having maximum skewness less than 0.86 for better simulation result. The growth rate parameter has been fixed at 1.2. Unsteady Reynolds-average Navier-Stokes (URANS) equations are used for the turbulent flow field. SST k-ω, a two-equation turbulence model is considered for the closure of turbulent flow field [26-28]. For Validation of our results, we considered the experimental results performed by Castelli et al. [11]. The comparison between these cases is shown in table 1.

Table 1: Geometrical characteristics

	Type of aerofoil	No. of blades	Diameter of rotor, D(m)	Chord of the blade	Solidity
Present work	NACA0017	3	0.7	0.1	0.425
Castelli et al.[11]	NACA0021	3	1.03	0.858	0.5